Streaming Data Analysis
Introduction

Emanuele Della Valle

prof @ Politecnico di Milano

founder @ Quantia Consulting

founder & CRO @ motus ml

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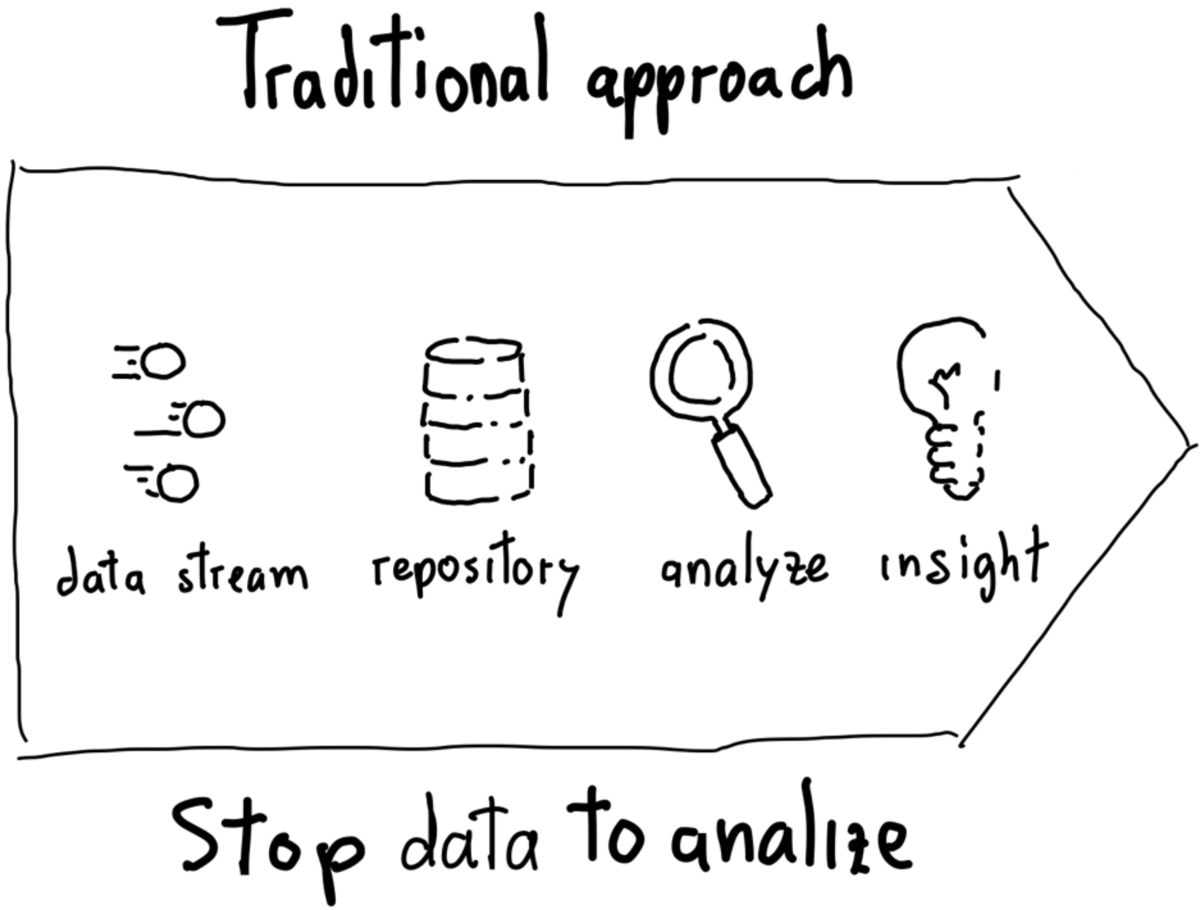
The vision

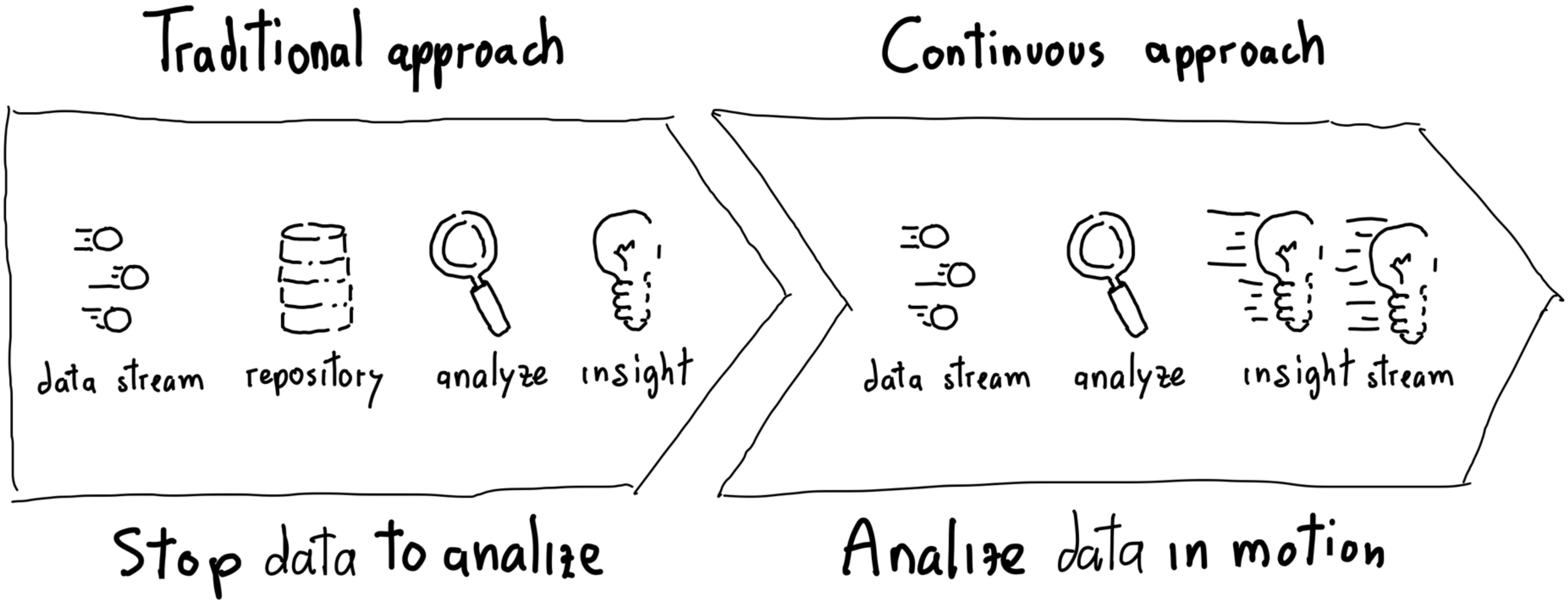
Continuity matters!

Continuity matters!

Traditional approach vs …



Traditional vs. Continuous approach



Traditional vs. Continuous approach

Monitoring each workout

Monitoring each step

Stop data to analize

Analize data in motion

Traditional vs. Continuous approach

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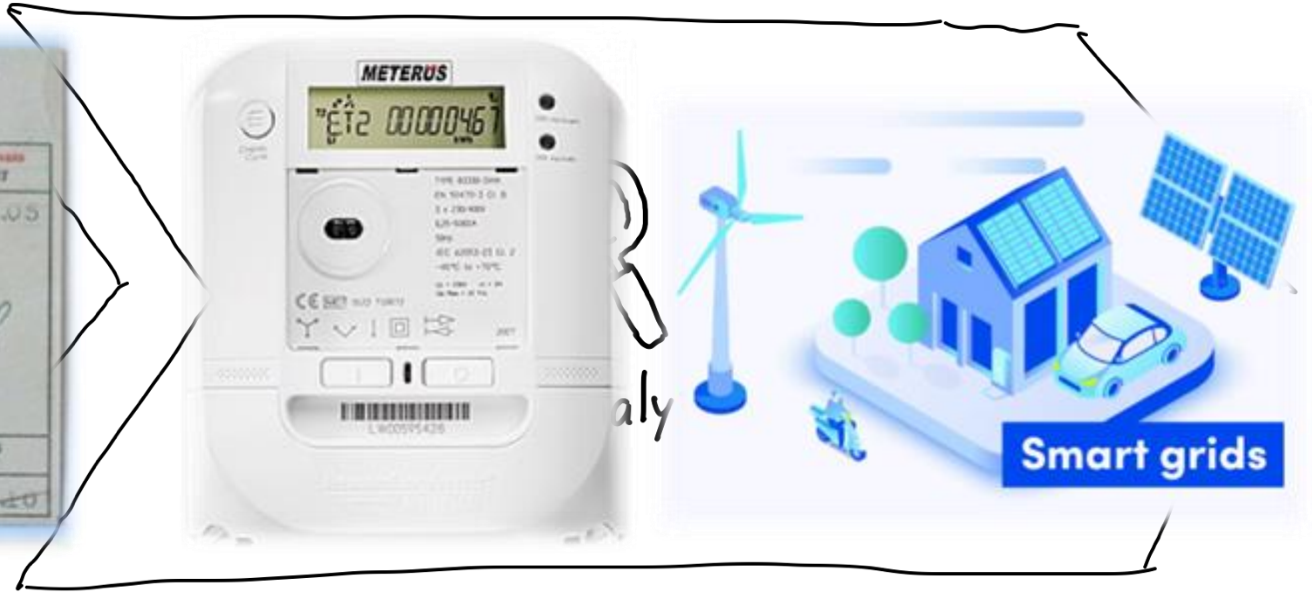
Disconnected electricity meters to bill consumption

Stop data to analize

Connected smart meters to enable smart-grid services



Analize data in motion

Traditional vs. Continuous approach

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Periodic exhaust gas inspection to authorize circulation

Stop data to analize

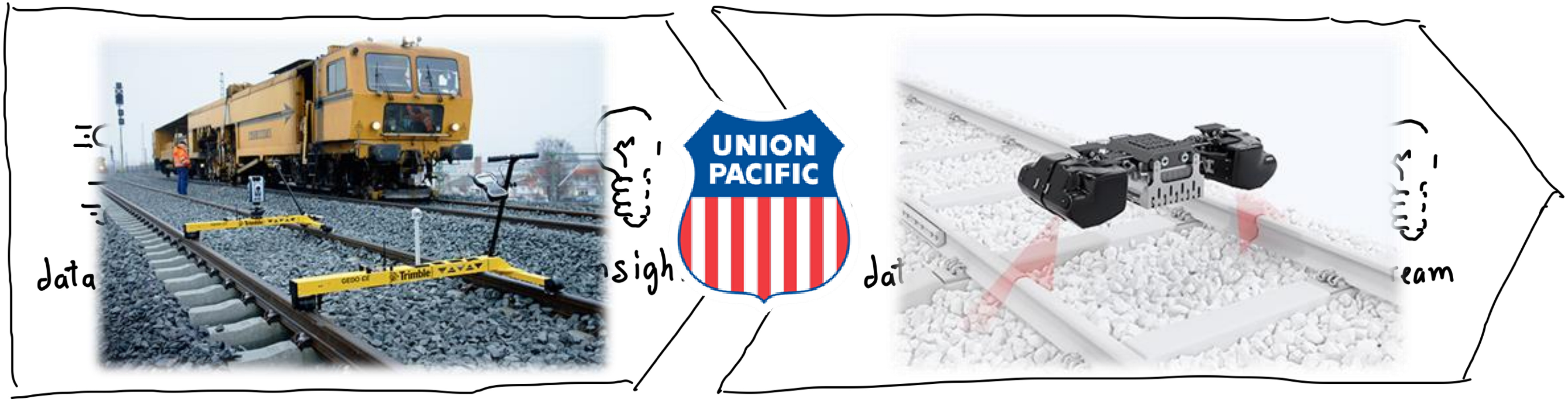
Lambda sensor to support environmentally friendly driving

Analize data in motion

Traditional vs. Continuous approach

Periodic quality checks
to ensure operations' safety

Continuous quality checks
to optimize maintenance



Stop data to analize

Analize data in motion

Guess one more!

Enabling Continuity

The key ingredients

1. Sensor and actuators
2. Connectivity
3. Streaming Data Engineering
4. Streaming Data Science

The key ingredients

1. **Sensor** and **actuators** (as sources and sinks, not covered in this course)
2. Connectivity
3. Streaming Data Engineering
4. Streaming Data Science

Sensors & actuators
The intuition

We need to **empower computers** with their own means of gathering information, **so they can see**, **hear** and **smell** the world for themselves, in all its random glory.

[…] sensor technology enable computers to **observe**, **identify** and **understand** the world - **without** the limitations of **human**-entered data.

-- Kevin Ashton, the brander of "The Internet of Things"

started life as the title of a presentation I made at Procter & Gamble (P&G) in 1999

https://www.rfidjournal.com/expert-views/that-internet-of-things-thing/73881/

Sensors
Eyes using cameras

Color Cameras

Fixed focus | auto focus | optical image stabilization

Color+Depth Cameras

Full info: https://stimulant.com/depth-sensor-shootout-2/

Actuators
Lights to display

Screens

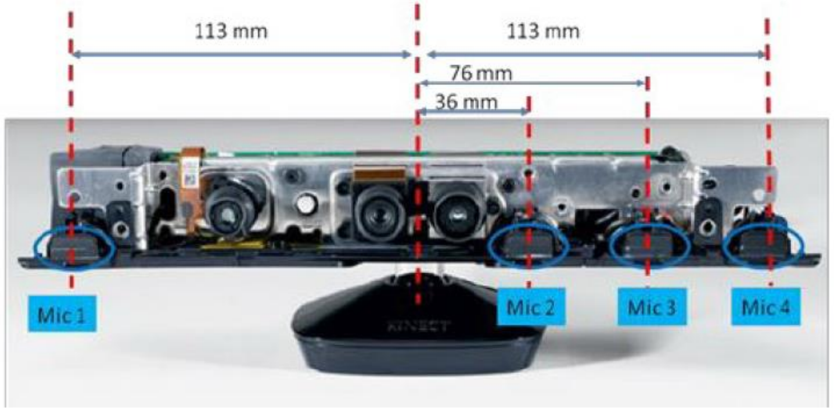
Projections

Holograms

Lights

Ears using microphones

Kinect

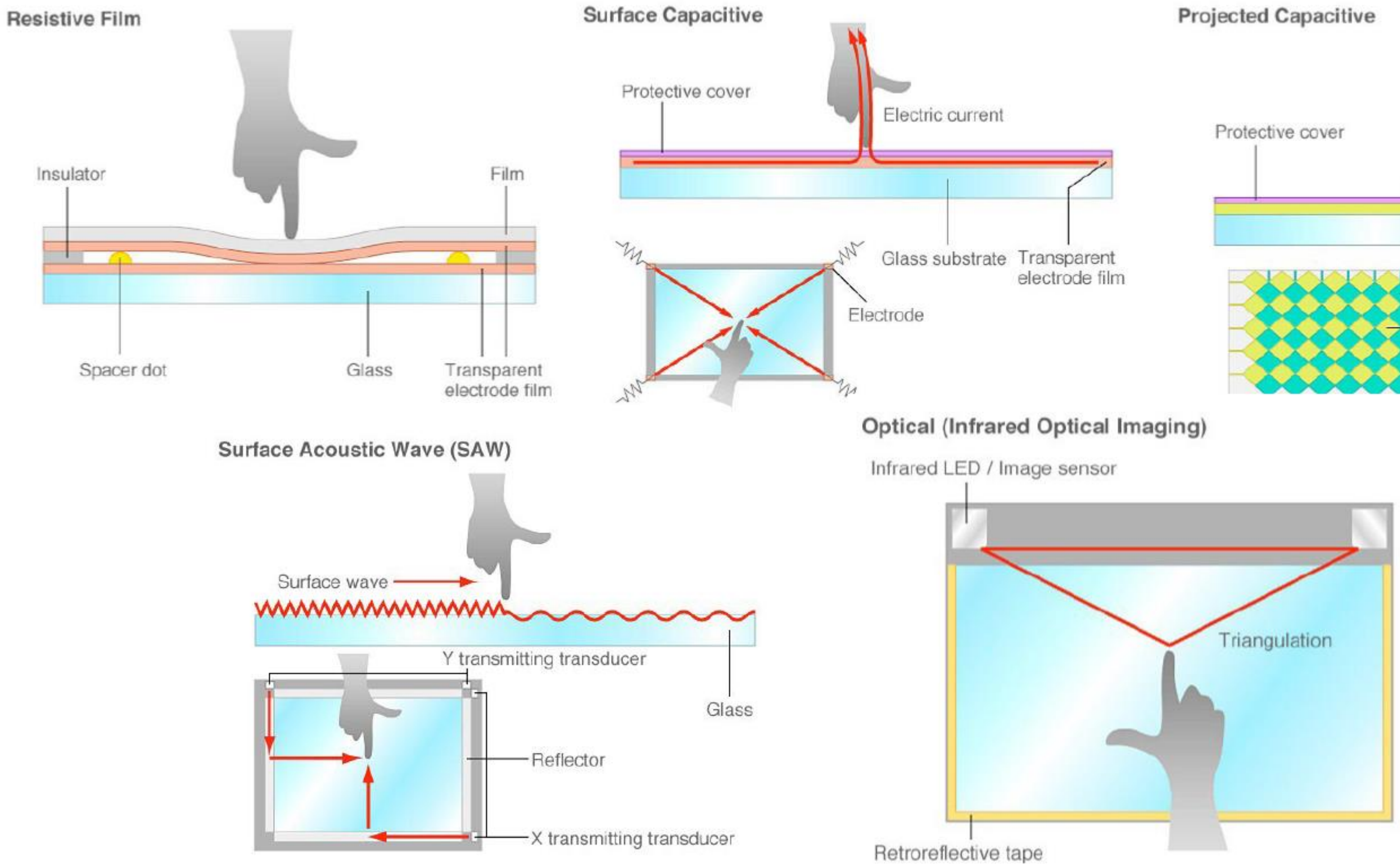


Microphone array

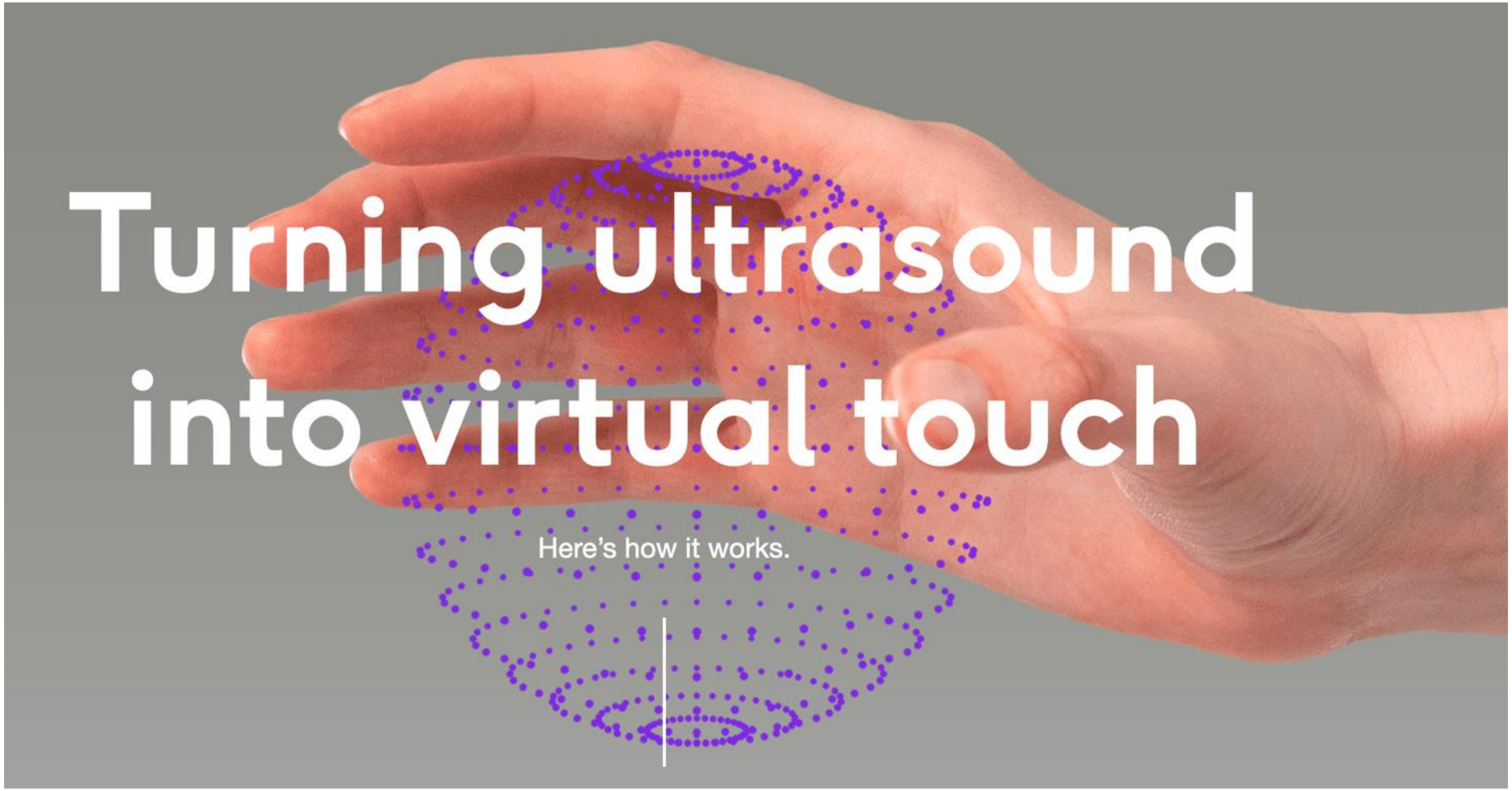
Microphone

Actuators
Speakers to speak

Sensors
Skin to feel touches



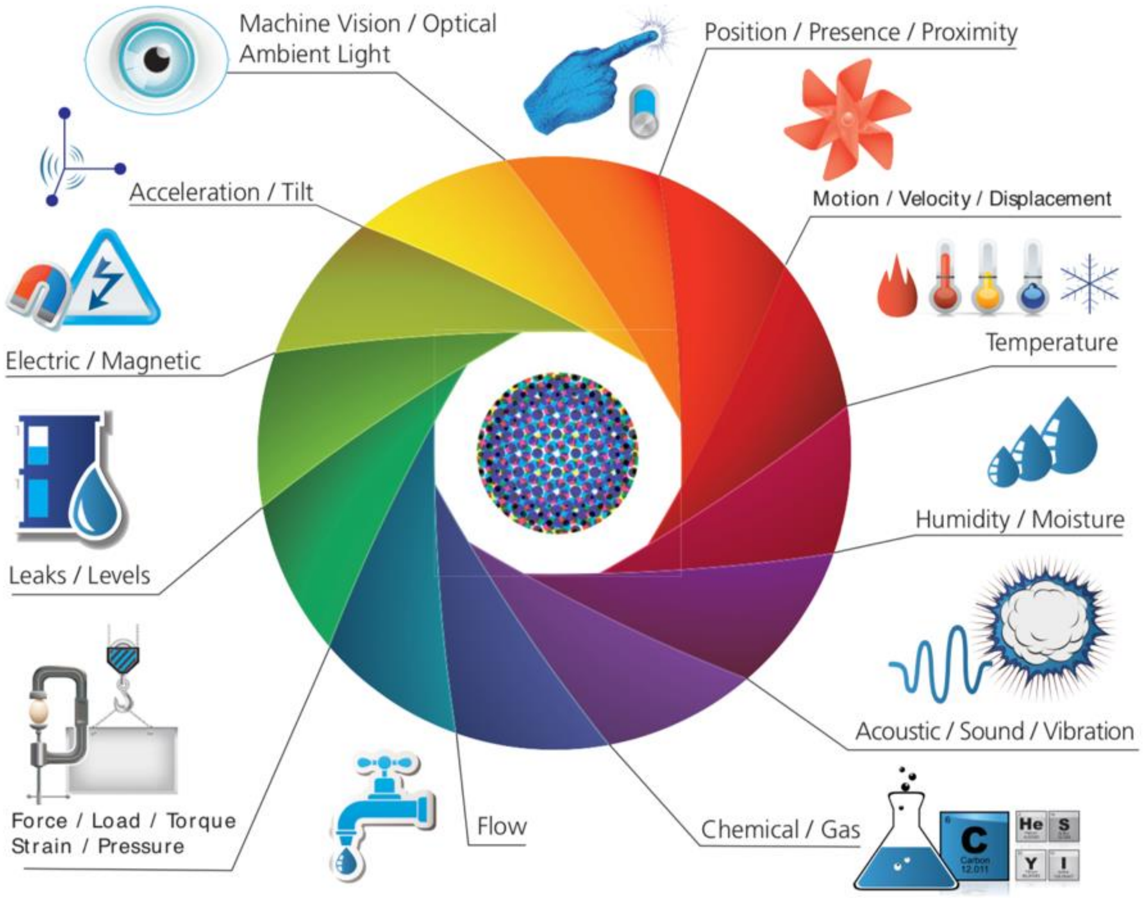
Actuators
Generate tactile sensations



[src: https://www.ultraleap.com/haptics/]

Sensors

… an entire digital nervous system …



[src: https://www.postscapes.com/iot-sensors-actuators/]

Sensors
… that gets smarter and smarter

[src: https://youtu.be/Cgxsv1riJhI]

Actuators
Stimulate other senses

Heating lamps

Fan

Bubble machine

Or cooling systems...

Fog machine

Actuators
Operating unmanned

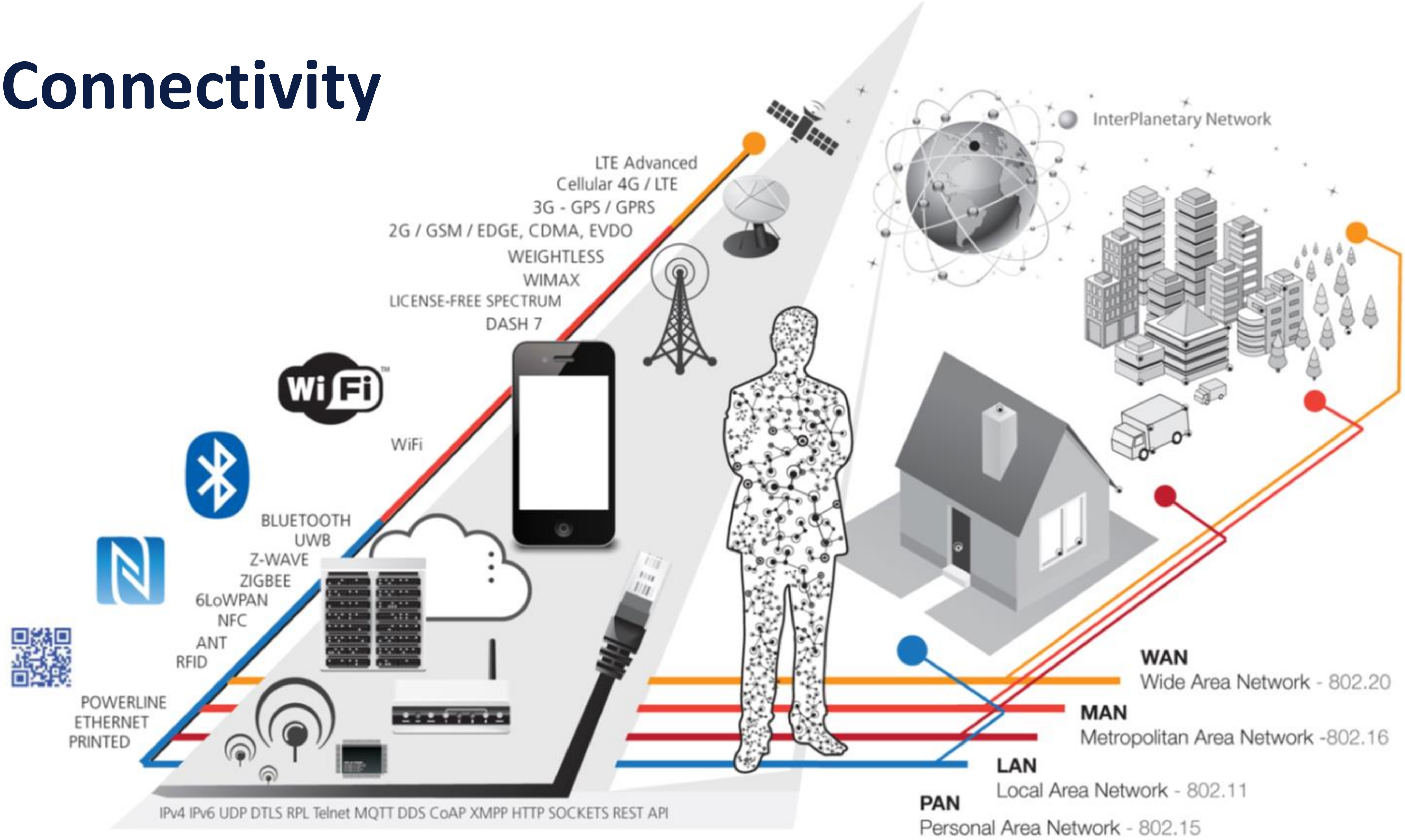
[src: https://youtu.be/QJGlh76YYpU]

[src: https://youtu.be/IzaaSIEDg7s]

The key ingredients

1. Sensor and actuators
2. **Connectivity** (as enabling and constraining factor, not covered in the course)
3. Streaming Data Engineering
4. Streaming Data Science

Connectivity



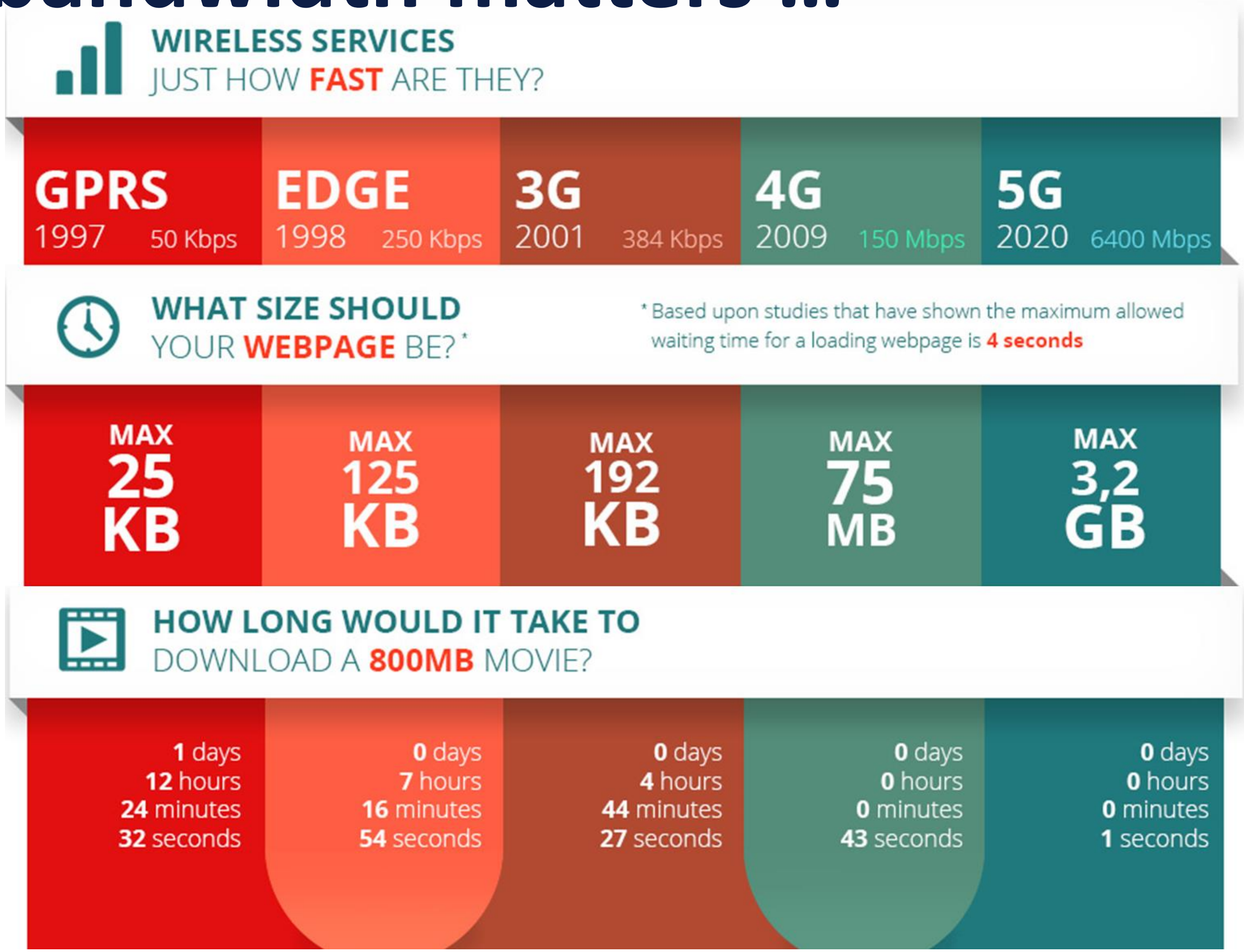
[src: https://www.postscapes.com/]

Connectivity
Distance and use case matters …

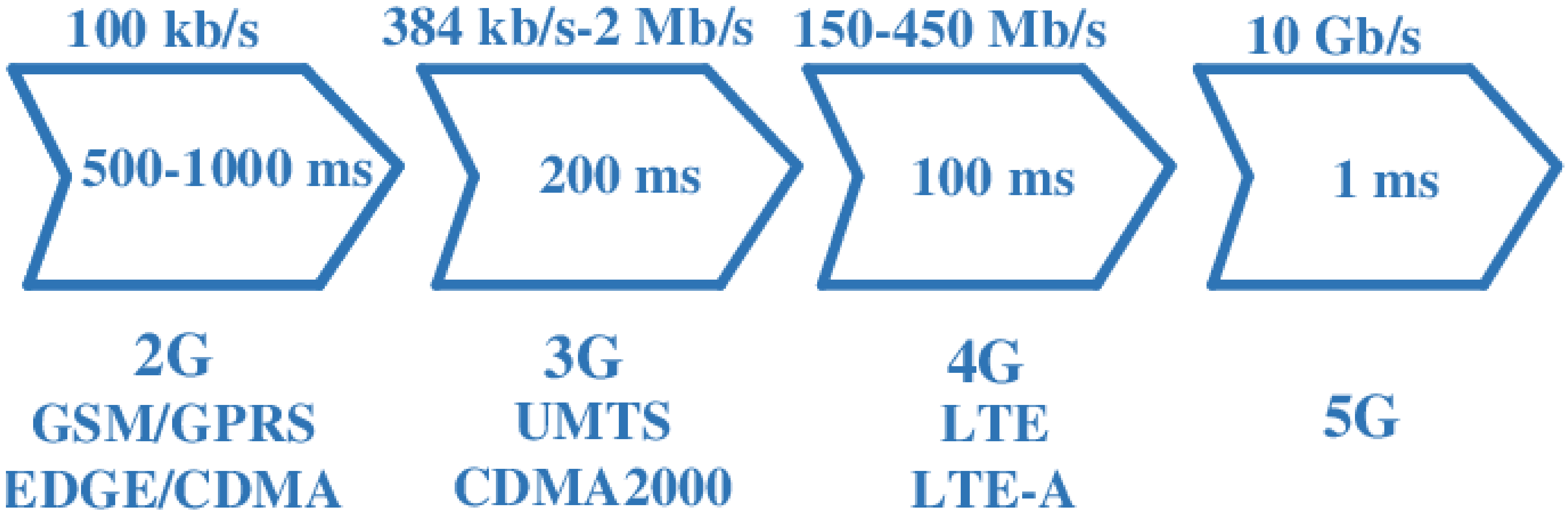
	Short Range	Local Area	Wide Area
Range (typical)	< 10m/30ft	< 100m/300ft	Outdoor (Km/Miles)
Content distribution Focus on high data rates	Bluetooth	WiFi, WiFi Direct, IEEE 802.11ax, .11ah: HaLow	GSM, lte, 5G
Sense & control Low energy/long battery life	Bluetooth SMART	zigbee, ULE Alliance	GPRS, NB-IoT
Proprietary solutions	ANT+	Sub-GHz, enocean	SIGFOX, LoRa, INGENU, WEIGHTLESS
Typical applications	**Personal appliances** (wrist band, smart watch, step counter, keyboard, mouse, pointer, etc.)	**Indoor networks** (internet, email, phone, security, energy management, home monitoring, etc.)	**Outdoor networks** (phone, chat, internet, smart city, industry 4.0, agriculture, smart logistics, etc.)

Connectivity

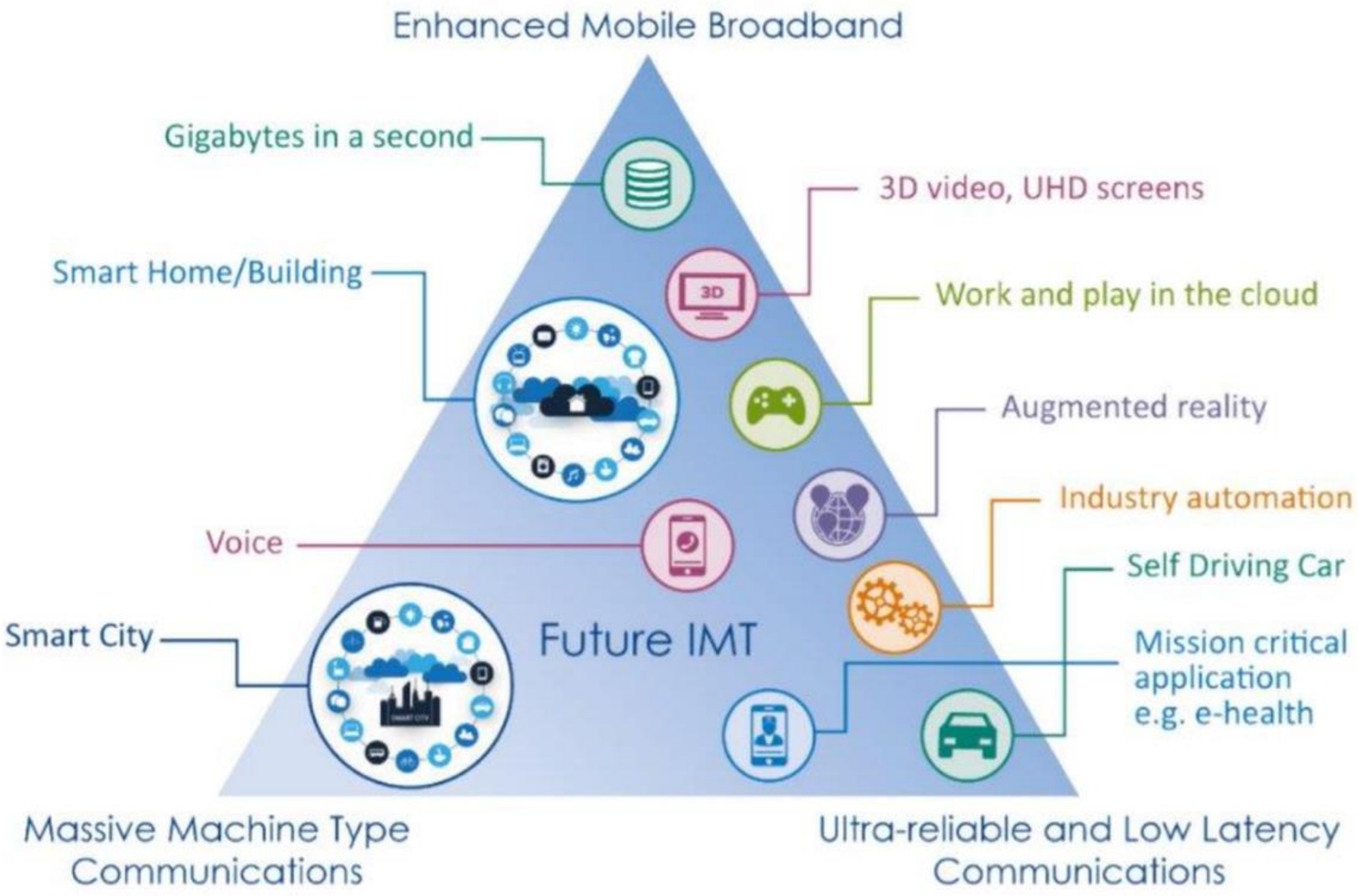
bandwidth matters …



Connectivity
latency matters …



Connectivity
… but beware of trade-off



The key ingredients

1. Sensor and actuators
2. Connectivity
3. **Streaming Data Engineering** (core topic of this course)
4. Streaming Data Science
5. Apps used by people in their processes

Taming *Myriads of tiny flows that you can collect* with **Event-based Systems**

Image by Pexels from Pixabay

Taming
Continuous massive flows
than you cannot stop
with
Data Stream Mng. Systems


Taming
Continuous numerous flows that
can turn into a torrent
with
Complex Event Processing

Taming

the forming of an immense delta made of myriads of flows of any size and speed with

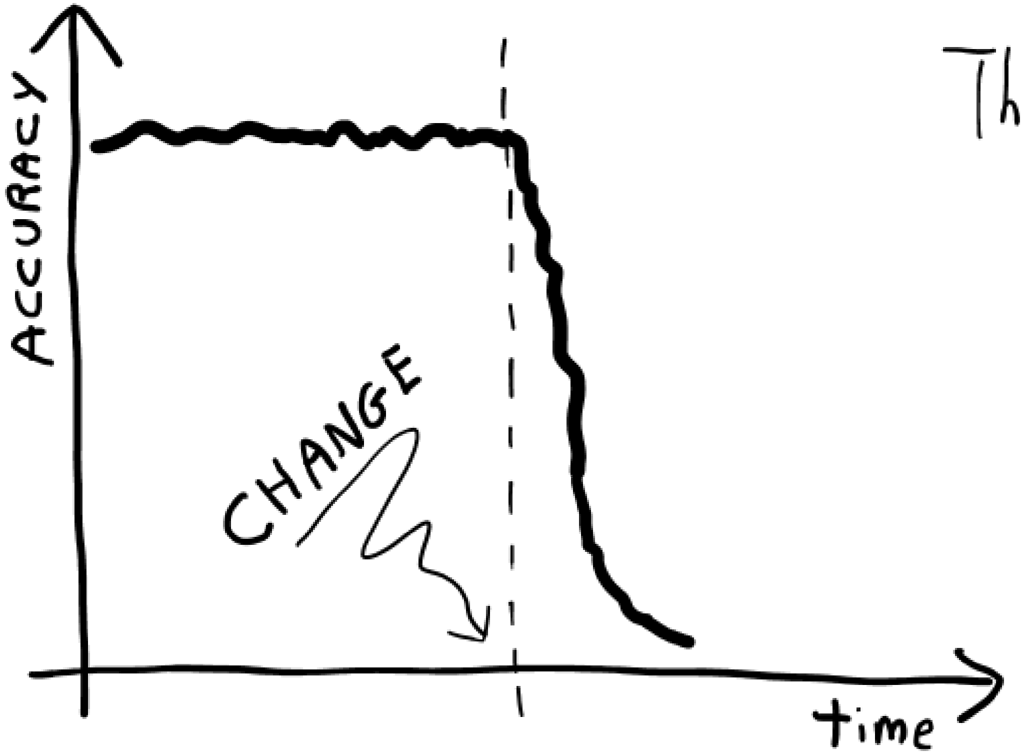
Event-driven Architecture

Image by Pexels from Pixabay

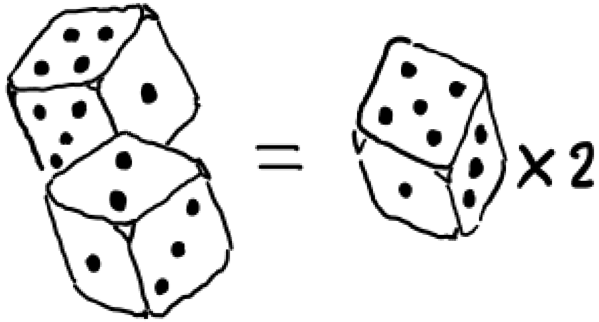
The key ingredients

1. Sensor and actuators
2. Connectivity
3. Streaming Data Engineering
4. **Streaming Data Science** (core topic of this course)

Changes cause ML models to lose accuracy



The **assumption**: data is
independent &
identically
distributed



It does **not** hold

Example of changes in natural language

- The word **Virus** in the **2010s** referred 55% of the time to **computers**, **suddenly** in **2020**, it referred 85% of the time to **medicine**
- **Mouse** in the **70s** referred 95% of the time to the **animal** slowly in the **90s** it started to refer **also to the device** and in the **2000s** it referred 85% of the time to the **device**
- **Turkey** in **July every year** refers 60% of the time to the **animal** in **November**, it refers 70% of the time to the **food**
- The word **selfie before the 2010s practically did not exist**

Example of changes in computer vision

- **Sudden changes** in the **acquisition device**, e.g., an x-ray machine 100 years ago vs a modern one
- **Slow changes** in the **status of the device**, e.g., lenses getting dirty
- **Seasonal changes** in the **environment**, e.g., road signals on sunny summer days vs. snowy winter ones
- **Recurring changes** in the **object**, e.g., players wearing different uniforms in different matches at home or away
- **Changes** can introduce **new objects**, e.g., the appearance of smartphones

Other examples of changes

- Churn reasons for customers of services
- Movies to recommend on IPTVs
- Survival rate in medicine
- …

Explaining the past and forecast the future of a continuous flow of data without assuming data independence with

Time Series Analytics

Learning one data at a time
from a continuous flow of data
without assuming identically distributed data
with
Streaming Machine Learning

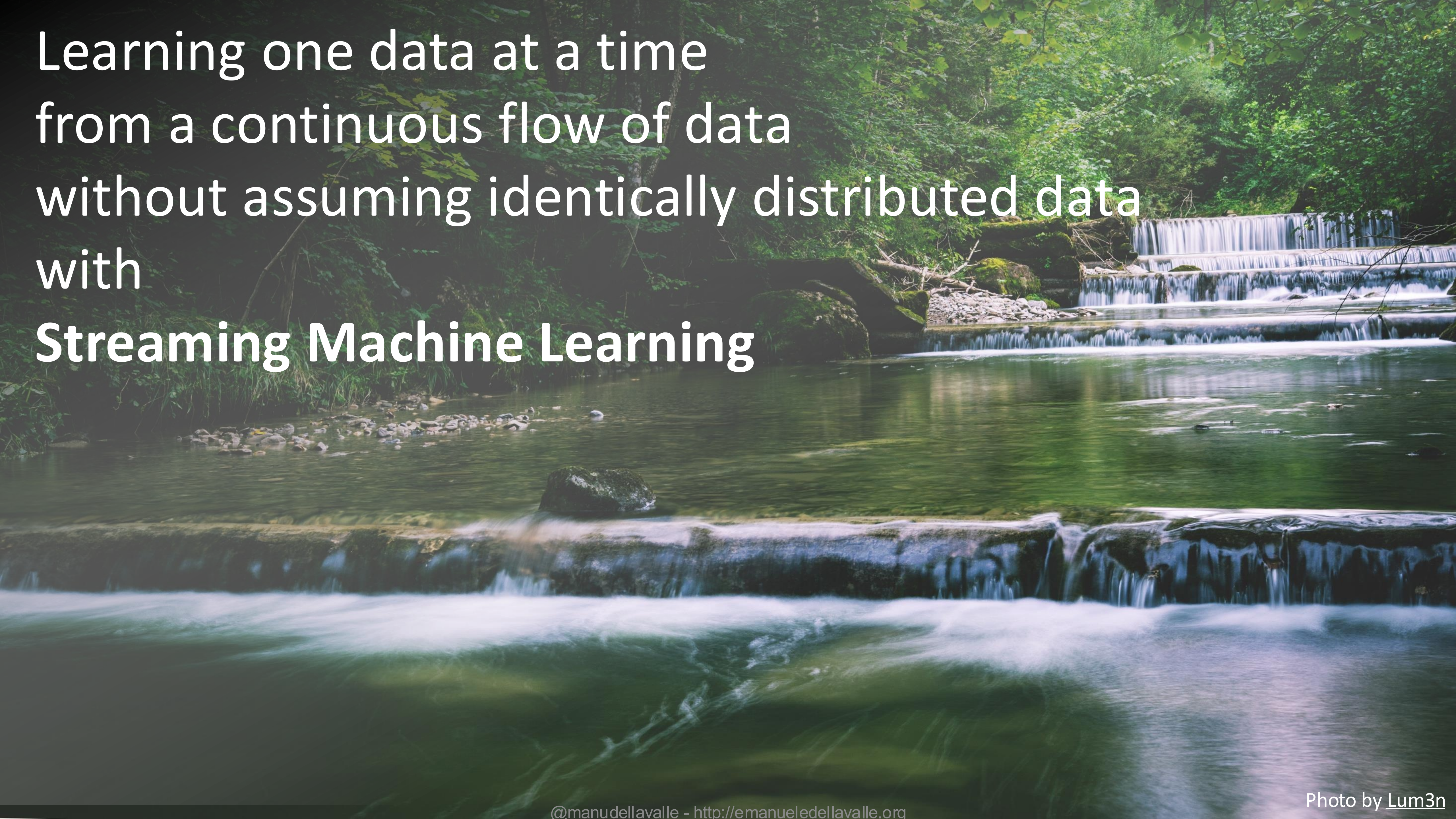
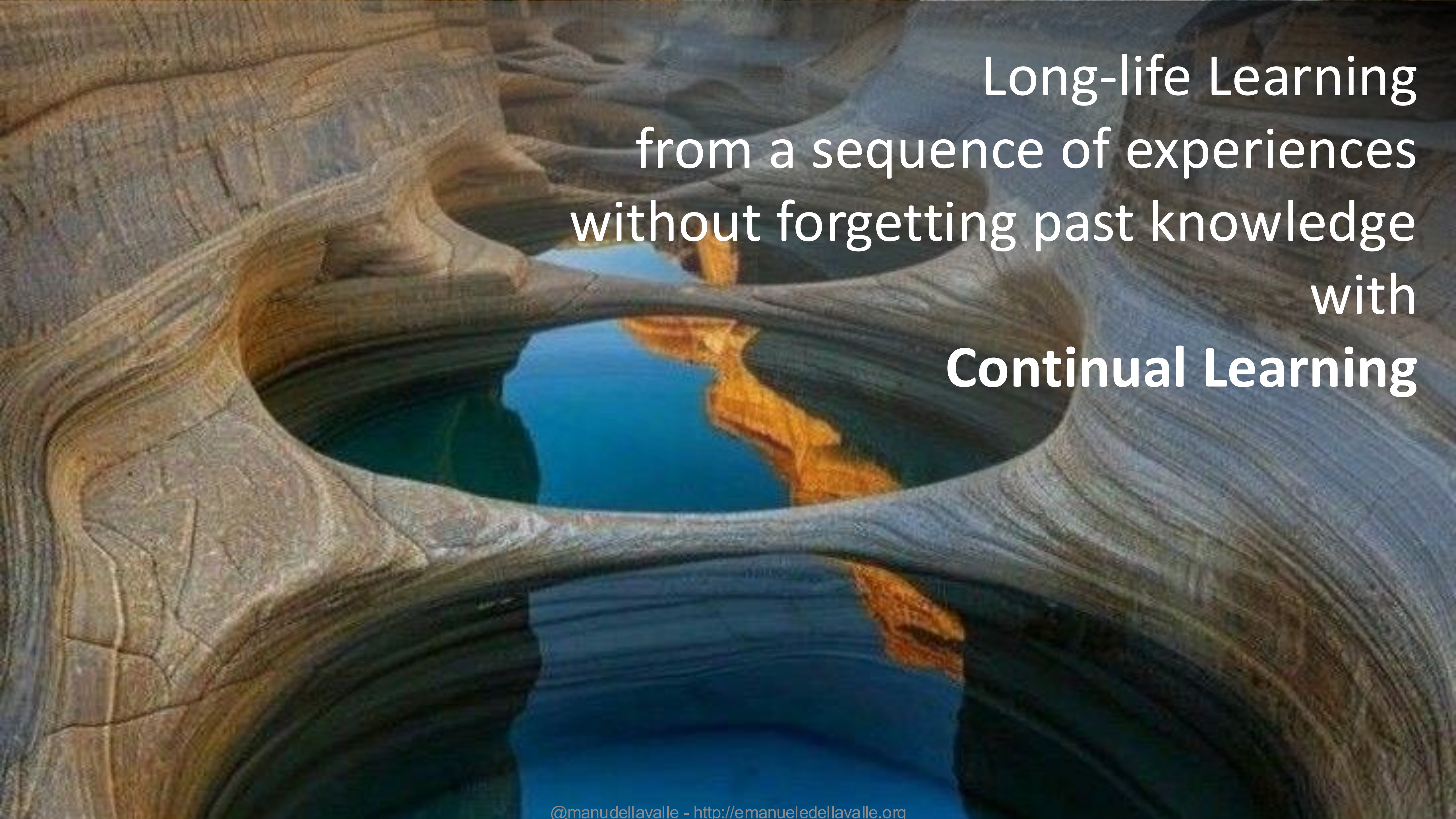
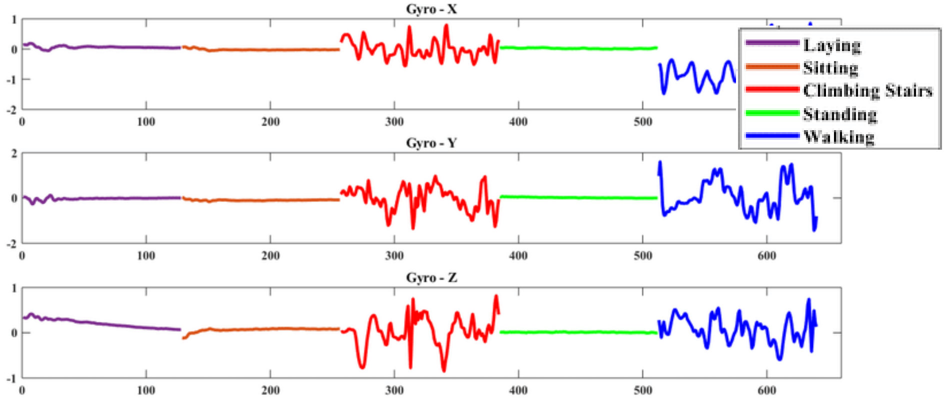


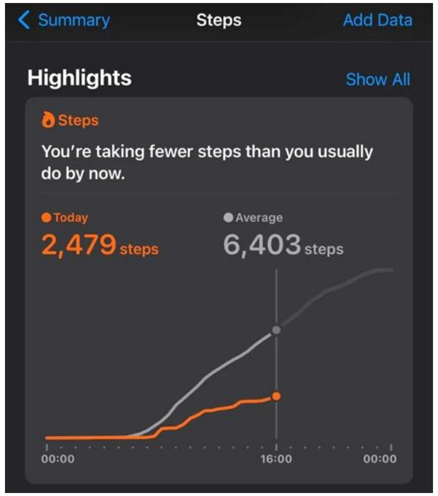
Photo by Lum3n


Long-life Learning
from a sequence of experiences
without forgetting past knowledge
with
Continual Learning

Cooking Streaming Data Analytics Apps
Steps' Analysis

- Sensor: gyroscopes and accelerometers
- Connectivity: on-chip
- Streaming Data Engineering:
  - identify steps over time
  - group steps by day
  - calculate the average (μ), the dispersion (σ), and the current (x) cumulative sum every quarter of an hour
- Streaming Data Science
  - Calculate z-score = $|x - \mu \ \sigma|$
- Actuators: notify user when z-score > 2





Cooking Streaming Data Analytics Apps
Learning Thermostats

- Sensor: temperature, humidity, proximity, presence, ambient light, users' positions, and weather forecasts
- Connectivity: Wi-Fi, BLE, OpenTherm
- Streaming Data Engineering: time-series DB
- Streaming Data Science: time-series analysis
- Actuators: 2 relays supporting a load of 3A at 230 V A

[src: https://store.google.com/product/nest_learning_thermostat_3rd_gen_specs]

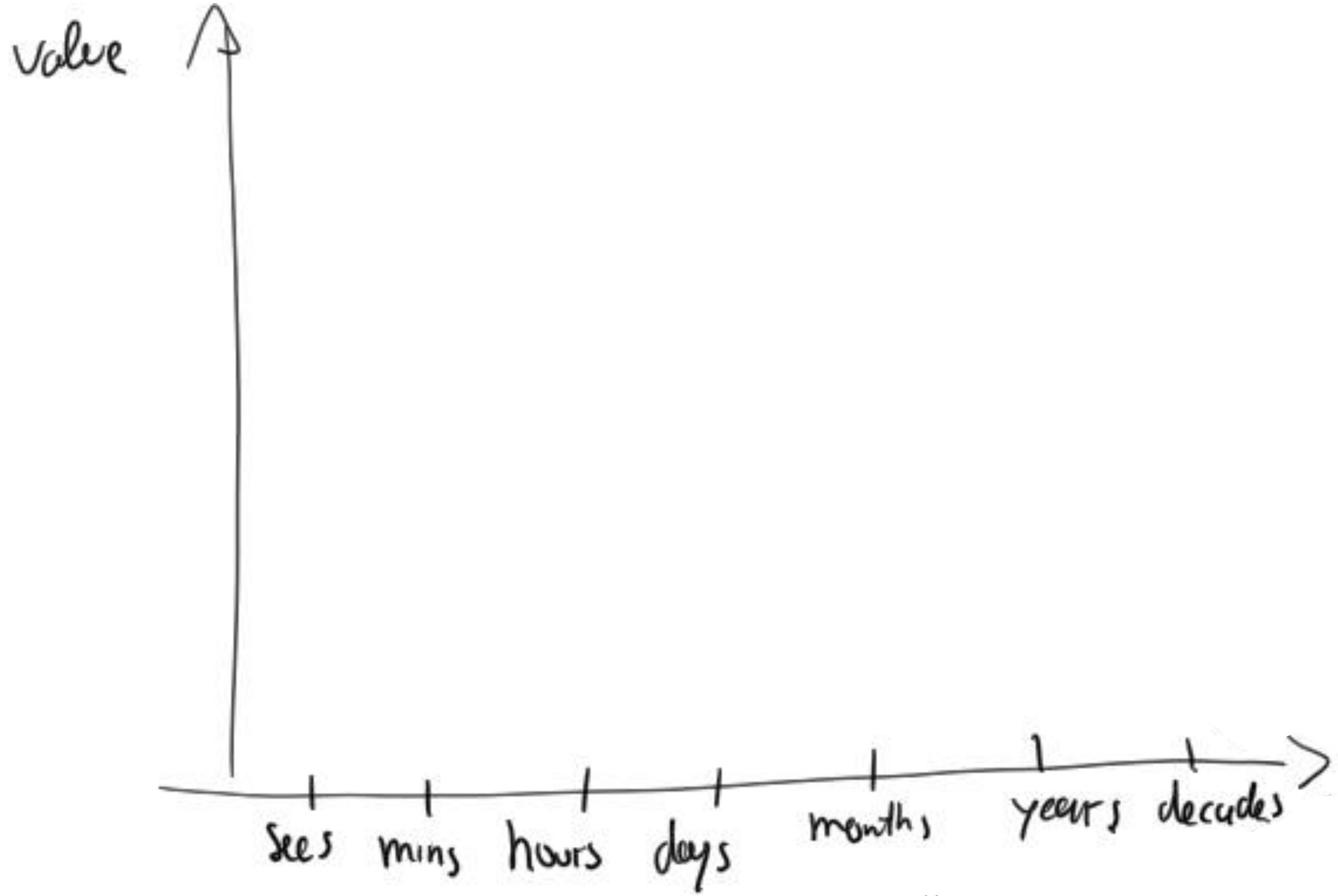
Name them all …

[src: https://youtu.be/v2kV6pgJxuo]

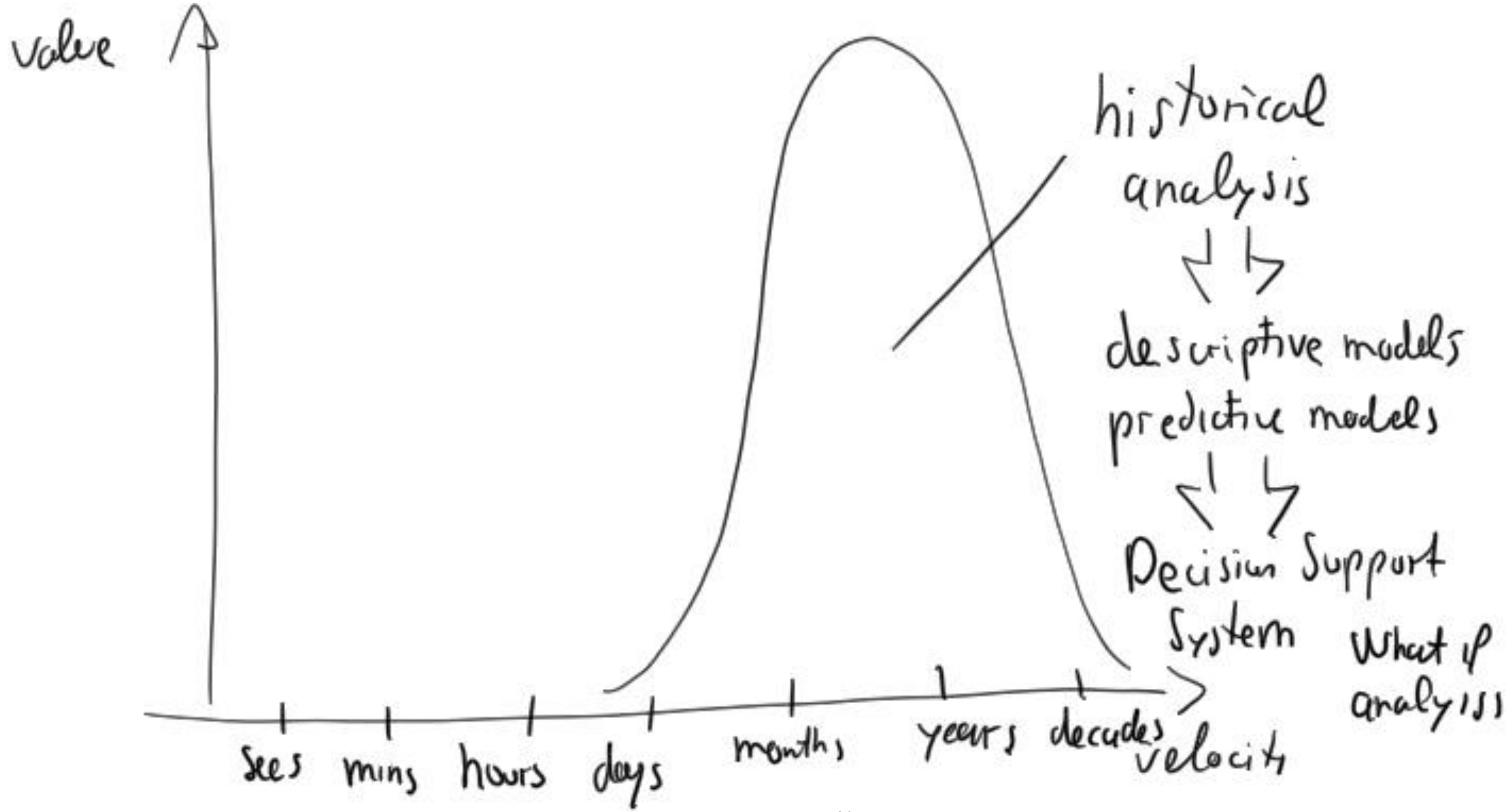
A final thought:

Value is about time!

Value as a function of time



Traditional value creation



A new opportunity

Hours? Minutes? Seconds? Milliseconds?

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Real-time?

~~Real~~-time

Right-time!

Act in time!!

Reactive!!!

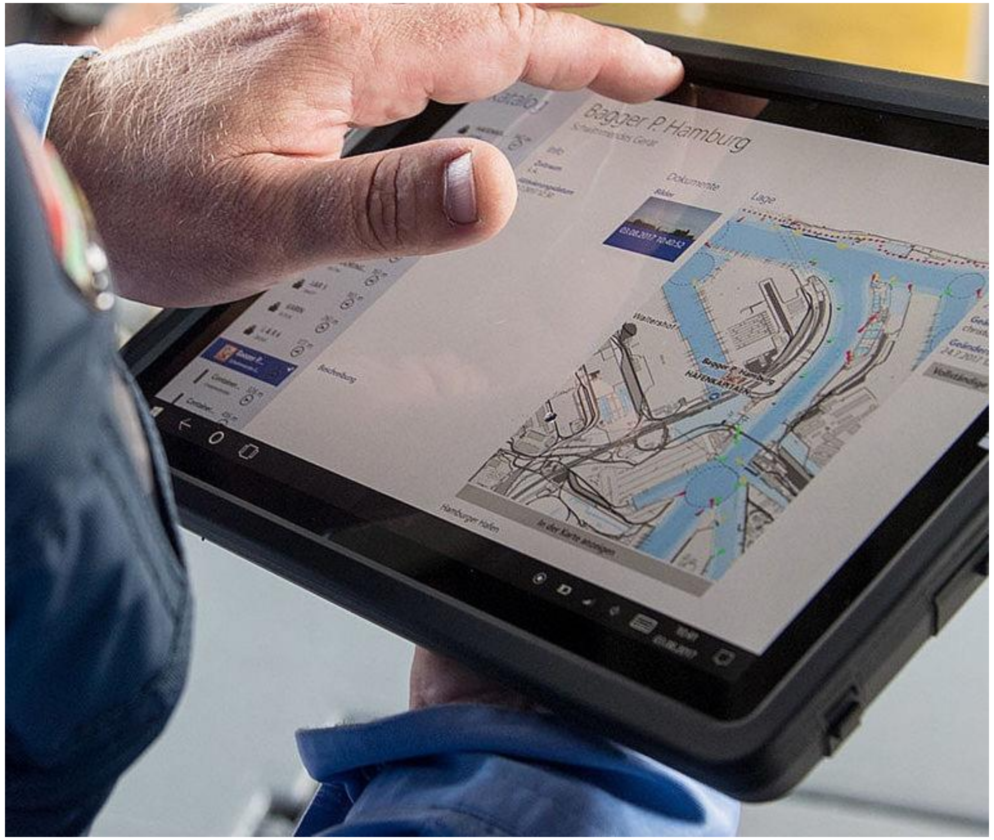
Continuous streaming value creation

Continuous streaming value creation

- reduce costs
- improve efficiency
- create innovative products
- new revenue streams

Intelligent Port

- Employing innovative technology to increase efficiency, effectiveness, and security by making logistics more environmentally sustainable, economically efficient, and capable of handling increasing traffic
- E.g., Hamburg Port Authority's Smart Port
  https://www.hamburg-port-authority.de/en/hpa-360/smartport/



Equipment as a service

- Pioneered in 1997 by Rolls-Royce, which lets airlines pay for their engines based on the number of flight hours. It is now a common business model in many industrial settings.

- E.g., Heller's insurance
  https://www.heller.biz/en/services/insurances/

Fruit-Picking Drones

- flying robots use artificial intelligence and machine learning to detect ripe apples before pulling the fruit from the tree
- E.g., Tevel
  https://www.tevel-tech.com/

[src: https://youtu.be/xzOf4-WaXzE]

:-D

[src: https://youtu.be/uaeADiepfXk]

Conclusion

Streaming Data Analytics is about

data **IN** time

processed **OVER** time

returning results **ON** time

Thesis offering

On-Board Data Processing and AI in the Space Cloud

Thales Alenia Space – Italy, Rome

Thesis objectives

- Contribute to the development of innovative solutions for the *Space Cloud*,
- Analyze enabling architectures and technologies (AI/ML on-board, edge computing, resource management)
- Explore application scenarios related to space missions, telecommunications, and Earth Observation

Activities

- Literature review on Space Cloud and related technologies.
- Analysis of requirements/KPI for software/hardware components for a Space Cloud
- Selection of alternative software/hardware components
- Prototype implementation
- Evaluation w.r.t. requirements and KPI

Credits

- The slides about sensors, actuators, and connectivity are partially based on those Matteo Valoriani presented in “Advanced User Interfaces: IoT & Mixed Reality” for the Interaction Design Laboratory course offered by Politecnico di Milano – Design Department

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